Total No. of Questions : 8 | | Total No. of Printed Pages : 3

Paper Code : 21307

F-407

B.C.A. (Second Semester) Examination, 2023

New Course

Paper No. BCA-N-202

DISCRETE MATHEMATICS

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Explain proposition and propositional form conditional.

   (b) Explain conjunction and disjunction with example.

2. (a) Simplify the following Boolean function

   (i) $\overline{A}B\overline{C} + ABC + \overline{A}BC + AB\overline{C}$

   (ii) $\overline{A}B + AB + B\overline{C}$

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   (b) Draw k-map

   $F(A,B,C) = \Sigma(2,3,4,6,7)$

3. (a) Give an example of a graph that is regular, but not complete, with each vertex having degree three.

   (b) Explain trees and its properties.

4. (a) Let $f:A \rightarrow B$ and $g:B \rightarrow C$ be functions. Show that if gof is onto then g is onto.

   (b) If A has m elements and B has n elements, how many functions are there from A to B?

5. (a) Explain min-terms and max-terms with example.

   (b) Explain AND, OR NOT operators.

6. (a) Prove that relation R defined on any non-void set A as $(a,b) \in R \Leftrightarrow a \geq b$ is partial order relation.

   (b) Explain equivalence relation with example.

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7. (a) What is graph? Explain finite and infinite graphs with example.

(b) Explain incidence matrix with example.

8. Write short notes on any **four** of the following :

(i) Boolean postulates

(ii) De Morgan's theorem

(iii) Connected graphs

(iv) Truth tables

(v) Inverse function

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